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EDUCATION

PhD., Biomedical Informatics, 2024 - Present
The Ohio State University, Columbus, OH, USA. (Qin Ma and Anjun Ma)
M.S., Biomedical, 2024
Zhengzhou University, Henan, Zhengzhou, China. (Ying Peng and Xia Xue)
B.S., Software Engineering, 2021
Zhengzhou University, Henan, Zhengzhou, China.

PROFESSIONAL EXPERIENCE

08/2024-present, **Graduate Research Assistant**
Department of Biomedical Informatics at The Ohio State University, OH

- Analyze bulk-level, single-cell, and spatial multi-omics datasets to derive meaningful insights for immuno-oncology research.
- Developed deep learning frameworks to enable cross-species analysis of single-cell and spatial omics data, providing new opportunities to investigate conserved and divergent molecular mechanisms across organisms.
- To establish a centralized AI-driven knowledgebase for integrating single-cell and spatial omics data across neurodegenerative diseases (ssKIND).

JOURNAL PUBLICATIONS

Full list: https://scholar.google.com/citations?user=kKMLVpUAAAAJ&hl=en
First and co-first author (*)

1. Liu, B.*, **Wu, W.***, Liu, C., Feng, P., Liu, S., Gong, S., Li, Y., Li, Y., Mi, J., Zheng, P., Wen, H., Xue, X., & Mi, Y. (2026). Multi-omic analysis of putative cellular senescence-associated genes on the risk of pan-cancer. Molecular Carcinogenesis. https://doi.org/10.1002/mc.70170

2. **Wu, W.***, Liu, B. *, Zhang, Q. #, Zhang, X., Feng, P., Jia, Y., & Xue, X. (2025). Heterogeneity and efficacy of immunotherapy in multiple cancer: insights from a meta-analysis. Biological Procedures Online, 27(1), 17.
3. Si, Y*., **Wu, W.***, Xue, X., Sun, X., Qin, Y., Li, Y., ... & Zheng, P. (2023). The evolution of SARS-CoV-2 and the COVID-19 pandemic. PeerJ, 11, e15990.
4. **Wu, W.**, Wang, Y., Xu, S., & Yan, K. (2020, September). Sfnn: Semantic features fusion neural network for multimodal sentiment analysis. In 2020 5th International Conference on Automation, Control and Robotics Engineering (CACRE) (pp. 661-665). IEEE.
5. Liu, S.*, **Wu, W.***, Ge, T.*, Zheng, P., Sun, X., Wang, J., Liu, B., Wen, H., Zhang, X., Zhu, Y., Hou, J., Zhang, Y., Mi, Y., & Xue, X. Integrated ATAC-seq and RNA-seq analysis reveals key transcriptional regulators driving intestinal metaplasia development in human organoids (Under Revision).
6. **Wu W**, Kim TY, Xu J, Cheng H, Wang C, …, Cheng F, Fu H, Ma A, Ma Q. An AI-powered single-cell and spatial omics ecosystem for neurodegenerative disease (Under Revision).
7. **Wu W***, Fu J*, Miller JG, Xiao K, Zhu J, Wang X, Zimmerman KA, Huang X, Li L, Li Z, Ma Q, Ma A. Toward computational preclinical models for advancing human health (in preparation).

Contributing author

1. Miller, S.J., Zhong, H., **Wu, W**., Cordova, A.M., Yashchenko, M.E., Yashchenko, A., Li, Z., Jafree, D.J., Zimmerman, C.N., DeVette, C.I. and Do, V., 2026. A cross model single-cell atlas reveals conserved involvement of osteopontin in polycystic kidney disease. JCI insight.
2. Wang, X., Duan, M., Su, P.L., Li, J., Krull, J., Jin, J., Chen, H., Sun, Y., **Wu, W**., He, K. and Carpenter, R., 2026. Deep-learning-enabled multi-omics analyses for prediction of future metastasis in cancer. Nature Communications.
3. Lin, D., Atkinson, J.R., **Wu, W.**, Chen, M., Jayasekara, S., Segal, B.M., Ma, A. and Gao, J., 2026. A self-limiting, TREM2-dependent anabolic program confers microglial resilience in Alzheimer's disease. bioRxiv.
4. Wang, Y., Ma, A., Song, N.J., Shannon, A.E., Amankwah, Y.S., Chen, X., **Wu, W**., Wang, Z., Saadey, A.A., Yousif, A. and Ghosh, G., 2025. Proteotoxic stress response drives T cell exhaustion and immune evasion. Nature.
5. Dong, H., Liang, C., Zhang, J., **Wu, W.**, Kumar, N., Liu, Z., Sun, Y., Liao, Z., Cheng, X., Yu, Y. et al., 2025. O-GlcNAc transferase plays dual antiviral roles by integrating innate immunity and lipid metabolism. Nature Communications.
6. Jiang, Y., Wang, S., Feng, S., Wang, C., **Wu, W**., Huang, X., Ma, Q., Wang, J. and Ma, A., 2024. scGNN+: Adapting ChatGPT for Seamless Tutorial and Code Optimization. bioRxiv.
7. Ye, S., Li, C., Zhao, R. and **Wu, W**., 2019. NOAA-LSTM: A new method of dialect identification. In Artificial Intelligence and Security: 5th International Conference, ICAIS 2019, New York, NY, USA, July 26-28, 2019, Proceedings, Part I 5 (pp. 16-26). Springer International Publishing.

TOOL DEVELOPMENT

- **ssKIND:** ssKIND: A Single-Cell and Spatial Omics Resource and AI Platform for Neurodegenerative Diseases (https://bmblx.bmi.osumc.edu/sskind/)

JOURNAL REVIEWER:

2025, **Heliyon, Guest Reviewer**

PRESENTATIONS

2026, ICIBM at Buffalo, NY ssKIND: A Single-Cell and Spatial Omics Resource and AI Platform for Neurodegenerative Diseases

CONFERENCE POSTERS

2026, ICIBM at Buffalo, NY ssKIND: A Single-Cell and Spatial Omics Resource and AI Platform for Neurodegenerative Diseases

MENTORING

Undergraduate Intern

Juile Zhu (B.S., Applied Mathematics in Neuroscience Physics Track, OSU) 2025/08 – current

Arya Wang (B.S., Applied Mathematics in Physics Track, OSU) 2025/09 – current

High school student intern

Konar, Suvam (now B.S. student, Carnegie Mellon University) 2026/06 - 2026/08

MEMBERSHIPS

- International Society for Computational Biology **(ISCB)** 2025 - Now

AWARDS

- Second place in the multimodal emotion recognition track of the 'KDDI' AI algorithm competition (Rank:2/456).
- China University Computer Challenge "Network Technology Challenge", Second Prize in Central China Region.
- ACM Programming Competition, Zhengzhou University, First Prize.
- "Coretronic Cup" Future Car Human-Machine Interaction Design Competition, 7 th (Top 0.1%)
- "Challenge Cup" National Competition of Extracurricular Academic Science and Technology Works for University Students, First Prize in Henan Province.

SKILLS

- **Bioinformatics & Data Analysis**: Next-generation sequencing, proteomics, single-cell multi-omics data analysis, high-performance computing (R, Python, HPC systems)
- **Software Development**: Java Boot Spring, Vue, Flask, Sql.
- **Machine Learning & AI**: Pytorch.
- **Data Visualization**: Creation of impactful visualizations (Adobe Illustrator, ggplot2, BioRender)